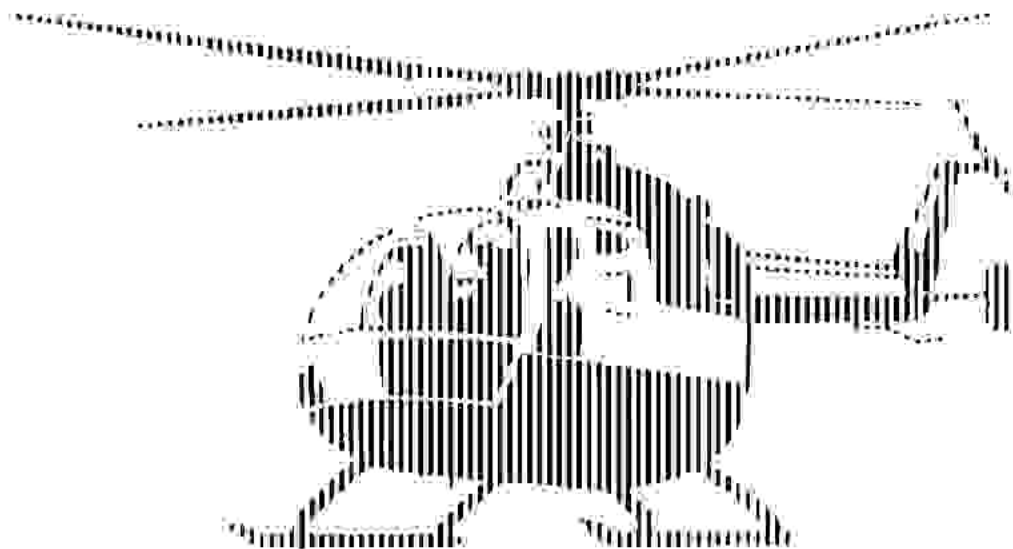


BO 105 CB/CBS

APPROVED ROTORCRAFT

FLIGHT MANUAL



Valid for Model BO 105 C Variants: **CB, CB-2, CDN-B, CB-4, and CDN-B-4**
and Model BO 105 S Variants: **CBS, CBS-2, CDN-BS, CBS-4, and CDN-BS-4**

Registration No. _____

Serial No. _____

LBA Type Certificate No. 3025 (3053 for prefix P model numbers)

THIS MANUAL MUST BE CARRIED IN THE HELICOPTER AT ALL TIMES.

IT IS THE OPERATOR'S RESPONSIBILITY TO MAINTAIN THIS MANUAL IN A
CURRENT STATUS IN ACCORDANCE WITH THE LIST OF EFFECTIVE PAGES

Fourth Issue — NOVEMBER 23, 1993

Approved by:

Luftfahrt-Bundesamt
Braunschweig

Date: 23.11.93

Published by:

EUROCOPTER DEUTSCHLAND GmbH
81663 München

APPROVING AUTHORITIES

The approved Model BO 105 C and S Variants included in this Flight Manual are listed below together with the appropriate approving authority.

VARIANTS	APPROVING AUTHORITY
CB, CB-4, CBS, CBS-4	LUFTFAHRT-BUNDESAMT (LBA) This manual is required by LBA Type Certificate Data Sheet No. 3025. NOTE BO 105 helicopter models having prefix P (e.g. P-BO 105) are manufactured under license by: Philippine Aerospace Development Corporation (PADC) Hangar No. 3, MIA-Avenue, Pasay, Metro Manila, Philippines For prefix P model helicopters, this manual is required by LBA Type Certificate Data Sheet No. 3053.
CB-2, CB-4, CBS-2, CBS-4	FEDERAL AVIATION ADMINISTRATION (FAA) This Flight Manual, required by FAA Type Certificate Data Sheet No. H3EU, is FAA approved as of April 14, 1994.
CDN-B, CDN-B-4, CDN-BS, CDN-BS-4	TRANSPORT CANADA AIRWORTHINESS GROUP (TCAG) This manual, required by TCAG Aircraft Type Approval No. H-79, is TCAG approved as of September 2, 1994.
ALL	EUROPEAN AVIATION SAFETY AGENCY (EASA) Beginning with Revision No. 6.1, revisions to this flight manual are approved by EASA based on Regulation (EC) No. 1592/2002 of the European Parliament and of the Council of 15 July 2002.
ALL	EUROPEAN AVIATION SAFETY AGENCY (EASA) Beginning with Revision No. 8 of this Flight Manual, revisions are approved by EASA based on Regulation (EC) No. 216/2008 of the European Parliament and of the Council of 20 February 2008.

MANUAL CONTENTS

CERTIFICATING AUTHORITY APPROVED PORTION

SECTION 1	GENERAL
SECTION 2	LIMITATIONS
SECTION 3	EMERGENCY PROCEDURES
SECTION 4	NORMAL PROCEDURES
SECTION 5	PERFORMANCE DATA
SECTION 10	OPTIONAL EQUIPMENT
SECTION 11	SPECIAL OPERATIONS

MANUFACTURER'S DATA PORTION

SECTION 6	MASS AND BALANCE
SECTION 7	SYSTEM DESCRIPTION
SECTION 8	HANDLING, SERVICING, AND MAINTENANCE
SECTION 9	OPERATIONAL INFORMATION
SECTION 12	APPENDIX

LOG OF REVISIONS

FOURTH ISSUE		NOV 23, 1993	REVISION 9	NOV 14, 2012
			EASA Approval no.:	10042197
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REVISION	2	SEP 15, 1995	EASA Approval no.:	10042326
REVISION	3	MAY 23, 1997	REVISION 10.1	JUL 25, 2013
REVISION	4	JUN 30, 2000	REVISION 11	SEP 24, 2014
			EASA approval no.:	10050590
REVISION	5	FEB 27, 2003	REVISION 11.1	MAY 04, 2017
REVISION	6	APR 15, 2004	REVISION 12	(see entry below)
REVISION	6.1	MAR 22, 2005		
REVISION	7	OCT 11, 2005		
EASA approval no.:	R.C.01231			
REVISION	7.1	FEB 14, 2006		
REVISION	8	MAR 12, 2010		
EASA Approval no.:	10029232			

NOTE The Temporary Revision No. 1 "Deactivation of CABLE CUTTER pb and description (FMS 10-17)" has been incorporated into the Basic FLM with Revision No. 12.

Revision 12

Approved by EASA

Date: DEC 05, 2017

EASA approval no.: 10063966

RECORD OF REVISIONS

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RECORD OF REVISIONS

RECORD OF REVISIONS (Continued)

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LIST OF EFFECTIVE PAGES

NOTE N, R, or D indicate pages which are New, Revised or Deleted respectively. Remove and dispose of superseded pages, insert the latest revision pages and complete the Record of Revisions as necessary.

Page	Rev No.	Remarks/Effectivity	Page	Rev No.	Remarks/Effectivity
Cover	—		1-12	0	
Title	0		1-13	3	
R Approving Authorities	12		1-14	0	
Manual Contents	0		1-15	0	
R Log of Revisions	12		1-16	3	
Record of Revisions	—		1-17	3	
			1-18	3	
Section 1			Section 2		
R LEP 1 thru LEP 9/ (LEP 10 blank)	12	List of Effective Pages	2-i	8	
Log of Temporary Revisions	—	To be issued when applicable	2-ii	8	
Record of Temporary Revisions	—		2-iii/(2-iv blank)	8	
List of Effective Temporary Revision Pages	—	To be issued when applicable	2-1	0	
Service Bulletin Incorporation Index	2		2-2	3	
			2-3	0	Variants CB, CB-2, CDN-B, CBS, CBS-2 and CDN-B
			2-3	0	Variants CB-4 and CBS-4 Variants CB-4 and CBS-4
1-i	6.1		2-4	0	
1-ii	0		2-5	0	
1-1	0		2-6	0	
1-2	0		2-7	1	
1-3	0		2-8	8	
1-4	0		2-9	8	
1-5	6.1		2-10	8	
1-6	6.1		2-11	8	
1-7	0		2-12	8	
1-8	0	Model C Variants	2-13	8	
1-8	0	Model S Variants	2-14	8	
1-9	0		2-15	8	
1-10	0		2-16	9	
1-11	0		2-17	9	

Page	Rev No.	Remarks/ Effectivity	Page	Rev No.	Remarks/ Effectivity
2-18	8		3-28	0	
2-19	8		3-29	0	
2-20	8		3-30	0	
2-21	8		3-31	5	
2-22	8		3-32	0	
2-23	8		3-33	0	
2-24	8		3-34	0	
2-25	8		3-35	0	
2-26	8		3-36	0	
2-27/(2-28 blank)	8		3-37	0	
Section 3			3-38	3	
3-i	2		3-39	0	
3-ii	0		3-40	0	
3-1	2		3-41	0	
3-2	9		3-42	6	
3-3	0		3-43	0	
3-4	8		3-44	0	
3-5	0		3-45	0	
3-6	0		3-46	8	
3-7	0		3-47	0	
3-8	3		3-48	0	
3-9	3		3-49/(3-50 blank)	0	
3-10	0		Section 4		
3-11	0		4-i	10	
3-12	9		4-ii	0	
3-13	0		4-1	1	
3-14	2		4-2	1	
3-15	1		4-3	0	
3-16	0		4-4	10	
3-17/(3-18 blank)	2		R 4-5	12	
3-19	0		R 4-6	12	
3-20	0		R 4-7	12	
3-21	0		4-8	10	
3-22	0		4-9	10	
3-23	0		4-10	10	
3-24	0		4-11	3	
3-25	0		4-12	5	
3-26	0		4-13	5	
3-27	0		4-14	8	

Page	Rev No.	Remarks/Effectivity	Page	Rev No.	Remarks/Effectivity
4-14A/ (4-14B blank)	4		5-31	2	
4-15	11.1		5-32	2	
4-16	8		5-33	1	
4-17	8		5-34	1	
4-18	0		5-35/(5-36 blank)	1	
4-19	0		Section 6		
4-20	3		6-i/(6-ii blank)	0	
Section 5			6-1	3	
5-i	3		6-2	3	
5-ii	3		6-3	0	
5-1	0		6-4	3	Model C Variants
5-2	0		6-5	0	Model C Variants
5-3	3		6-6	0	Model C Variants
5-4	3		6-7/(6-8 blank)	2	Variants CB, CB-2 and CDN-B
5-5	3		6-7/(6-8 blank)	2	Variants CB-4 and CDN-B-4
5-6	3		6-4	3	Model S Variants
5-7	3		6-5	0	Model S Variants
5-8	3		6-6	2	Model S Variants
5-8A	3		6-7/(6-8 blank)	2	Variants CBS, CBS-2 and CDN-BS
5-8B	3		6-7/(6-8 blank)	2	Variants CBS-4 and CDN-BS-4
5-9	1		Form MBR-1	0	Front
5-10	5		Form MBR-1	0	Back
5-11	1		Form EL-1	0	Front
5-12	1		Form EL-1	0	Back
5-13	1		Section 7		
5-14	1		7-i	3	
5-15	1		7-ii	0	
5-16	1		7-iii	0	
5-17	1		7-iv	0	
5-18	1		7-1	0	
5-19	1		7-2	3	
5-20	1		7-3	0	Model C Variants
5-21	2		7-3	0	Model S Variants
5-22	1		7-4	0	
5-23/(5-24 blank)	1		7-5	0	
5-25/(5-26 blank)	2				
5-27/(5-28 blank)	1				
5-29/(5-30 blank)	1				

Page	Rev No.	Remarks/ Effectivity	Page	Rev No.	Remarks/ Effectivity
7-6	0		8-3	0	
7-7	0		8-4	0	
7-8	0		8-5	0	
7-9	0		8-6	0	
7-10	0		8-7	0	
7-11	0		8-8	0	
7-12	0		8-9/(8-10 blank)	0	
7-13	0		Section 9		
7-14	0		9-i	7.1	
7-15	0		9-ii	7.1	
7-16	0		9-1	0	
7-17	0		9-2	0	
7-18	2		9-3/(9-4 blank)	0	
7-19	2		9-5/(9-6 blank)	0	
7-20	9		9-7/(9-8 blank)	0	
7-21	0		9-9/(9-10 blank)	0	
7-22	0		9-11/(9-12 blank)	0	
7-23	0		9-13/(9-14 blank)	0	
7-24	8		9-15/(9-16 blank)	0	
7-25	0		9-17/(9-18 blank)	0	
7-26	0		9-19/(9-20 blank)	0	
7-27	0		9-21	0	
7-28	3		9-22	0	
7-28A/ (7-28B blank)	3		9-23	0	
7-29	0		9-24	0	
7-30	3		9-25	0	
7-31	0		9-26	0	
7-32	0		9-27/(9-28 blank)	0	
7-33	0		9-29/(9-30 blank)	0	
7-34	0		9-31/(9-32 blank)	0	
7-35	0		9-33	0	
7-36	2		9-34	0	
7-37	0		9-35	0	
7-38	0		9-36	7.1	
Section 8			Section 10		
8-i/(8-ii blank)	0		Log of Supple- ments 1	4	
8-1	2		Log of Supple- ments 2	4	
8-2	0				

Page	Rev No.	Remarks/Effectivity	Page	Rev No.	Remarks/Effectivity
Log of Supplements 3	4		10-12-2	11	
Log of Supplements 4	3		10-12-3	11	
Log of Supplements 5/(6 blank)	3		10-12-4	11	
10-1-1	0		10-12-5	11	
10-1-2	0		10-12-6	11	
10-1-3/(10-1-4 blank)	0		10-12-7	11	
10-2-1	0		10-12-8	11	
10-2-2	0		10-12-9	11	
10-3-1	2		10-12-10	11	
10-3-2	0		10-12-11	11	
10-4-1	0		10-12-12	11	
10-4-2	0		10-12-13	11	
10-5-1	0		10-12-14	11	
10-5-2	0		10-12-15	11	
10-5-3/(10-5-4 blank)	2		10-12-16	11	
10-6-1	0		10-12-17	11	
10-6-2	2		10-12-18	11	
10-6-3/(10-6-4 blank)	0		10-12-19	11	
10-7-1	0		10-12-20	11	
10-7-2	2		10-12-21	11	
10-8-1	0		(10-12-22 blank)		
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10-8-3/(10-8-4 blank)	2		10-13-2	0	
10-9-1	0		10-13-3/(10-13-4 blank)	0	
10-9-2	0		10-14-1	0	
10-10-1	0		10-14-2	4	
10-10-2	0		10-14-3/(10-14-4 blank)	2	
10-10-3	0		10-15-1	0	
10-10-4	0		10-15-2	0	
10-11-1	2		10-15-3	2	
10-11-2	0		10-15-4	0	
10-11-3	2		10-15-5/(10-15-6 blank)	0	
10-11-4	1		10-16-1	0	
10-12-1	11		10-16-2	2	
			10-17-1	0	
			10-17-2	0	
			10-17-3	0	

Page	Rev No.	Remarks/ Effectivity	Page	Rev No.	Remarks/ Effectivity
10-17-4	0		10-24-3	0	
R 10-17-5	12		10-24-4	0	
R 10-17-6	12		10-24-5	1	
R 10-17-7	12		10-24-6	2	
R 10-17-8	12		10-24-7	0	
R 10-17-9	12		10-24-8	0	
R 10-17-10	12		10-24-9	0	
R 10-17-11	12		10-24-10	1	
R 10-17-12	12		10-25-1	0	
R 10-17-13	12		10-25-2	0	
N 10-17-14	12		10-25-3	4	
10-18-1	0		10-25-4	2	
10-18-2	2		10-25-5	2	
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10-20-2	2		10-27-3	0	
10-21-1	0		10-27-4	0	
10-21-2	3		10-28-1	0	
10-21-3	0		10-28-2	2	
10-21-4	0		10-28-3	2	
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10-22-2	0		10-28-5/	0	
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10-23-2	2		10-29-5/	0	
10-23-3	2		(10-29-6 blank)		
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			(10-30-4 blank)		

Page	Rev No.	Remarks/ Effectivity	Page	Rev No.	Remarks/ Effectivity
10-31-1	0		10-37-4	0	
10-31-2	0		10-37-5	0	
10-31-3	2		10-37-6	2	
10-31-4	0		10-37-7	0	
10-32-1	0		10-37-8	2	
10-32-2	0		10-37-9	0	
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10-33-1	0		10-38-1	0	
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10-33-3	0		10-38-3	0	
10-33-4	0		10-38-4	2	
10-33-5	0		10-38-5	0	
10-33-6	0		10-38-6	0	
10-33-7	0		10-39-1	1	
10-33-8	2		10-39-2	1	
10-33-9	2		10-39-3	1	
10-33-10	0		10-39-4	2	
10-34-1	0		10-40-1	2	
10-34-2	0		10-40-2	0	
10-34-3	2		10-40-3	0	
10-34-4	2		10-40-4	0	
10-35-1	0		10-41-1	0	
10-35-2	0		10-41-2	0	
10-35-3	0		10-41-3	0	
10-35-4	2		10-41-4	0	
10-35-5	0		10-41-5	0	
10-35-6	0		10-41-6	0	
10-36-1	0		10-41-7	0	
10-36-2	3		10-41-8	2	
10-36-3	5		10-41-9	0	
10-36-4	1		10-41-10	0	
10-36-5	1		10-42-1	0	
10-36-6	0		10-42-2	0	
10-36-7	2		10-42-3	0	
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10-37-2	0		10-42-6	0	
10-37-3	0		10-42-7/ (10-42-8 blank)	2	

Page	Rev No.	Remarks/ Effectivity	Page	Rev No.	Remarks/ Effectivity
10-44-1	0		10-55-1	0	
10-44-2	0		10-55-2	2	
10-44-3	0		10-55-3	0	
10-44-4	2		10-55-4	0	
10-44-5	0		10-56-1	4	
10-44-6	0		10-56-2	4	
10-45-1	0		10-56-3	4	
10-45-2	0		10-56-4	4	
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10-45-4	2		10-56-6	4	
10-45-5	0		10-57-1	0	
10-45-6	0		10-57-2	0	
10-46-1	2		10-57-3/ (10-57-4 blank)	2	
10-46-2	5		10-60-1	2	
10-46-3/ (10-46-4 blank)	2		10-60-2	0	
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10-47-3	2		10-60-5	0	
10-47-4	0		10-60-6	0	
10-48-1	2		Section 11		
10-48-2	0		Log of Supple- ments 1/(2 blank)	5	<i>All Variants Ex- cept CB-4 and CBS-4</i>
10-48-3/ (10-48-4 blank)	0		Log of Supple- ments 1/(2 blank)	6.1	<i>Variants CB-4 and CBS-4</i>
10-49-1	2		11-1-1	0	
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10-51-2	0		11-2-2	5	
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10-52-2	3		11-2-6	0	
10-52-3/ (10-52-4 blank)	3		11-2-7/ (11-2-8 blank)	0	
10-54-1	3		11-3-1	0	
10-54-2	3		11-3-2	2	
10-54-3	3				
10-54-4	3				

Page	Rev No.	Remarks/Effectivity	Page	Rev No.	Remarks/Effectivity
11-3-3	0		11-4-28	1	
11-3-4	0		11-4-29	1	
11-4-i	5	<i>All Variants Except CB-4 and CBS-4</i>	11-4-30	1	
			11-4-31	1	
11-4-i	5	<i>Variants CB-4 and CBS-4</i>	11-4-32	1	
11-4-ii	1		11-4-33	1	
11-4-iii	1		11-4-34	1	
11-4-iv	1		11-4-35	1	
11-4-1	5	<i>All Variants Except CB-4 and CBS-4</i>	11-4-36	1	
			11-4-37/	1	
11-4-1	5	<i>Variants CB-4 and CBS-4</i>	(11-4-38 blank)		
11-4-2	2				
11-4-3	1				
11-4-4	0				
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11-4-8	1				
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11-4-10	2				
11-4-11	2				
11-4-12	2				
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11-4-14	1				
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11-4-16	0				
11-4-17	0				
11-4-18	0				
11-4-19/	0				
(11-4-20 blank)					
11-4-21	5				
11-4-22	1				
11-4-23	1				
11-4-24	1				
11-4-25	1				
11-4-26	1				
11-4-27	1				

RECORD OF TEMPORARY REVISIONS

[illegible]

SERVICE BULLETIN INCORPORATION INDEX

SB No.	Title/Description	FLM Sections Affected	FLM Revision Incorporated
60-31	Installation of an Engine Compartment Additional (Forced) Ventilation	2, 5, 9	0
60-37	Retrofit of Ram Air Ventilation for Oil Cooling System	2, 3, 7	0

SECTION 1

GENERAL

TABLE OF CONTENTS

	<u>Page</u>
1.1 GENERAL	1 - 1
1.2 ORGANIZATION OF THE MANUAL	1 - 1
1.2.1 Approved Data	1 - 1
1.2.2 Description of the Sections	1 - 1
1.2.3 Page Numbering	1 - 2
1.2.4 Flight Manual Supplements (FMS)	1 - 3
1.2.5 Flight Manual Appendices (FMA)	1 - 3
1.3 EFFECTIVITY STATEMENTS	1 - 4
1.3.1 Definition of Effectivity Identification	1 - 4
1.3.2 Method of Indicating Effectivity	1 - 4
1.3.3 Incorporating Pages Having Entire Page Effectivity Statements	1 - 5
1.4 REVISION SERVICE	1 - 5
1.4.1 Revisions	1 - 5
1.4.2 Minor Revisions	1 - 5
1.4.3 Temporary Revisions	1 - 5
1.4.4 Identifying Revised Material	1 - 5
1.5 FLIGHT MANUAL APPLICABILITY	1 - 6
1.6 GENERAL DESCRIPTION OF THE HELICOPTER	1 - 7
1.7 HELICOPTER DIMENSIONS	1 - 7
1.8 CONVERSION CHARTS	1 - 7
1.9 TERMINOLOGY AND DEFINITIONS OF TERMS	1 - 15
1.9.1 Warnings, Cautions and Notes	1 - 15
1.9.2 Use of Procedural Terms	1 - 15
1.10 ABBREVIATIONS AND SYMBOLS	1 - 16

LIST OF FIGURES

	<u>Page</u>
Fig. 1-1 Principal Dimensions	1 - 8
Fig. 1-2 Conversion Chart: Celsius - Fahrenheit	1 - 9
Fig. 1-3 Conversion Chart: Kilometers/Hour - Knots	1 - 10
Fig. 1-4 Conversion Chart: Meters/Second - Feet/Minute	1 - 11
Fig. 1-5 Conversion Chart: Kilograms - Pounds	1 - 12
Fig. 1-6 Conversion Chart: Liters - US Gal - Imp. Gal	1 - 13
Fig. 1-7 Conversion Chart: Hectopascals - Inches of Mercury	1 - 14

SECTION 1

GENERAL

1.1 GENERAL

This manual contains the information necessary for safe and efficient operation of the BO-105 helicopter.

The user is assumed to have flying experience, therefore elementary instructions and basic principles have been omitted.

1.2 ORGANIZATION OF THE MANUAL

This flight manual is largely organized in the standardized format recommended by the Helicopter Association International.

It is divided into an **approved part** consisting of sections 1 thru 5, 10 and 11, and into a **non-approved part** consisting of sections 6 thru 9 and 12.

1.2.1 Approved Data

This manual meets all certificating authority requirements for approved data.

1.2.2 Description of the Sections

The sections of the manual are fairly independent with each section beginning with its own table of contents.

Approved Part

SECTION 1 GENERAL

presents information of general interest to the pilot, basic helicopter data and conversion tables. In addition it provides definitions and explanations of symbols, abbreviations, and terminology used in the manual.

SECTION 2 LIMITATIONS

contains those limitations required by regulation or necessary for safe operation of the helicopter and approved by the regulatory authority.

SECTION 3 EMERGENCY AND MALFUNCTION PROCEDURES

contains the recommended procedures for coping with various types of emergencies, malfunctions or critical situations.

SECTION 4 NORMAL PROCEDURES

contains the recommended procedures for normal ground and air operation of the helicopter.

SECTION 5 PERFORMANCE DATA

contains airworthiness and performance information necessary for preflight and inflight mission planning.

SECTION 10 OPTIONAL EQUIPMENT

contains Flight Manual Supplements (FMS) providing operating instructions of approved optional equipment.

SECTION 11 SPECIAL OPERATIONS

contains Flight Manual Supplements (FMS) with instructions for special operations (e.g. IFR, operation with opened doors, etc.).

Non-approved Part**SECTION 6 MASS AND BALANCE**

contains the definitions for various mass and balance locations and the procedure for the determination of the center of gravity.

Appended to this section are the **Mass and Balance Record** form for maintaining a continuous record of changes in structure and equipment affecting the mass and balance, and the **Equipment List** form for the listing of optional equipment with data necessary for mass and balance computations.

SECTION 7 SYSTEM DESCRIPTION

contains a brief description of the helicopter, its systems and the various standard equipment with information considered most important to the flight crew.

SECTION 8 HANDLING, SERVICING, AND MAINTENANCE

contains servicing data, cleaning and care procedures as well as information for ground handling.

SECTION 9 OPERATIONAL INFORMATION

contains information for flight planning such as fuel consumption and max cruising speed diagrams.

SECTION 12 APPENDIX

contains Flight Manual Appendices (FMA) for non-approved optional equipment and special operations.

1.2.3 Page Numbering

The numbering of pages within each section consists of the section number or designation, a dash and the consecutive number of the page beginning with "1", e.g. for Section 3: 3 - 1, 3 - 2, etc.

Figures are likewise numbered consecutively by section, such as Fig. 3-1, Fig. 3-2, etc.

Exceptions:

- The numbering of the **Table of Contents** pages preceding each section in this manual consists of the section number, a dash and the consecutive Roman numeral (lower case) of the page, beginning with "i"; e.g. for Section 3: 3 - i, 3 - ii, etc.
- The page numbers of the flight manual supplements and appendices consist of the section number, a dash, the number of the supplement/appendix, a dash and the consecutive number of the page; e.g. 10 - 17 - 1, 10 - 17 - 2, etc.
- Figures within a flight manual supplement/appendix are numbered consecutively, such as Fig. 1, Fig. 2, etc.

The numbers of blank reversed pages are printed on the preceding page by using dual page numbering; e.g. 3 - 9/(3 - 10 blank).

If, at a later date, pages have to be added to the initial printing, the new pages may carry the number of the preceding page plus a letter suffix; e.g. 2 - 6A, 2 - 6B, etc.

1.2.4 Flight Manual Supplements (FMS)

Each FMS is self-contained and corresponds in its general arrangement to the basic flight manual, but only additional information or different data will be the subject of an FMS.

Each FMS, although complete in nature, shall therefore be used in conjunction with the basic flight manual.

A Log of Supplements is provided for each applicable section as an index listing the current supplements.

The manufacturer retains the right to convert optional equipment to standard equipment at any time as a product improvement program. FLM coverage of the converted optional equipment, however, will remain as an FMS in Section 10 and also as an optional equipment item entry in the Equipment List. An entry in the STD EQPT EFFECTIVITY column of the Log of Supplements is used to indicate the converted optional equipment by providing an effectivity statement of the affected helicopters.

1.2.5 Flight Manual Appendices (FMA)

Each FMA is self-contained and corresponds in its general arrangement to the basic flight manual, but only additional information or different data will be the subject of an FMA.

Each FMA, although complete in nature, shall therefore be used in conjunction with the approved flight manual.

When two or more appendices are issued, a Log of Appendices will be provided as an index listing the current appendices.

1.3 EFFECTIVITY STATEMENTS

1.3.1 Definition of Effectivity Identification

The contents of this manual applies to all the BO-105 helicopter model variants listed on the title page and defined in this section. However, certain portions of the manual may apply to only specific models, variants, serial numbered helicopters, etc. Therefore, an effectivity identification system is used to indicate where differences brought about by helicopter modifications, Service Bulletins, customer options, variations of the basic model helicopter, etc. occur within the manual.

This system comprises effectivity statements, identified by the word **EFFECTIVITY** followed by the definition of applicability. The applicability may be defined using:

- a specified model or variant type,
- an eight-digit numeric indicator,
- a Service Bulletin (SB) modification number, and/or
- a clear text definition.

The eight-digit numeric indicator begins with the four digits of the lowest assigned manufacturer's serial number, to indicate first effectivity, and ends with the four digits of the highest assigned manufacturer's serial number, to indicate last effectivity, of an unbroken sequence of assigned serial numbered helicopters. A hyphen is shown between the two numbers. Open ended effectivity is indicated by "9999" in the last effectivity, e.g. 0415-9999 indicates helicopter serial number S-415 and subsequent.

An SB effectivity is indicated as either "Before SB....." or "After SB....." meaning respectively before and after incorporation of the specific numbered Service Bulletin on the helicopter.

1.3.2 Method of Indicating Effectivity

The amount of manual coverage required by the differences brought about by model, variants or modification differences (effectivity) varies from small or minor (portions of text) to extensive (entire pages). Therefore, the effectivity statements may apply to portions of a page or an entire page.

Effectivity statements applying to portions of a page immediately precede and are located in the extreme left hand margin directly above the text to which they relate. When the defined applicability ends, an effectivity statement defined as "ALL" is used to mean that the information following applies again to all models, variants, etc. as listed on the FLM title page or, when applicable, as defined by the page effectivity statement.

Effectivity statements applying to the entire page content will be located at the bottom of the page adjacent to the page number.

1.3.3 Incorporating Pages Having Entire Page Effectivity Statements

Compare the effectivity statements of those pages having identical page numbers and determine which page(s) applies. Insert that page(s) in the FLM binder. Discard the page(s) that does not apply.

NOTE Do not discard pages having a page effectivity statement "After SB....." that at present may not apply to the helicopter unless however, it is absolutely certain that the stated Service Bulletin will never be incorporated.

1.4 REVISION SERVICE

This manual is kept up-to-date by revisions and temporary revisions.

1.4.1 Revisions

Revisions are issued periodically. They are printed on white paper and are incorporated into the manual in accordance with a "Letter of Transmittal" sheet which need not be inserted in the manual. Revisions are numbered consecutively beginning with the number 1.

1.4.2 Minor Revisions

The publisher of this Flight Manual (EUROCOPTER) is authorized by the certifying authorities (EASA) to publish revisions when necessary to correct minor errors (typing errors, misprints, etc.) or minor changes without the requirement for certification by the authorities. In this case, the procedures for a normal revision apply, however, the revision numbering is modified in that the number of the last revision approved by the certifying authorities is used followed by a decimal point and then the minor revision number, e.g. Rev. 6.1, Rev. 6.2, etc.

1.4.3 Temporary Revisions

Temporary revisions are provided to transmit information between revisions. They are printed on yellow paper and are accompanied by a "Letter of Transmittal" sheet and an updated "List of Effective Temporary Revisions" page.

Temporary revisions are numbered consecutively beginning with the number 1; e.g. Temp. Rev. 1, Temp. Rev. 2, etc.

1.4.4 Identifying Revised Material

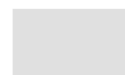
Changes (except as noted below) to the text and tables (including new material on added pages) are indicated by a vertical line in the outer margin.

Change symbols will not be shown for:

- Introductory material.
- Blank space resulting from the deletion of text, or an illustration or a part of an illustration, or table.

- Correction of minor inaccuracies, such as spelling, punctuation, relocation of material, etc, unless such correction changes the meaning of instructive information and procedures.

Changes to illustrations (except diagrams and schematics) are indicated by a miniature pointing hand. A vertical line next to changed text and callouts on illustrations may be used in lieu of a pointing hand. Shading and screening are used for diagrams and schematics to highlight the area containing the changed information. Extensively changed presentations may be indicated by a screen border around the affected area.



1.5 FLIGHT MANUAL APPLICABILITY

This flight manual is applicable for the following listed BO-105 models and variants (including those having model number prefix "P", e.g. P-BO 105 S).

BO-105 Model	Variants and Characteristics	Approval Authority
C	CB Max GM: 2400 kg (5291 lb) Engines: Allison 250-C20B TOP per engine: 257 kW (345 SHP) Main transmission: ZF FS 72 B	LBA
	CB-2 - Same as Variant CB	FAA
	CDN-B - Same as Variant CB	TCAG
	CB-4 Max GM: 2500 kg (5512 lb) Engines: Allison 250-C20B TOP per engine: 257 kW (345 SHP) Main transmission: ZF FS 72 B	LBA FAA
	CDN-B-4 - Same as Variant CB-4	TCAG
S	CBS (BO-105 CB Stretched) Max GM: 2400 kg (5291 lb) Engines: Allison 250-C20B TOP per engine: 257 kW (345 SHP) Main transmission: ZF FS 72 B	LBA
	CBS-2 - Same as Variant CBS	FAA
	CDN-BS - Same as Variant CBS	TCAG
	CBS-4 Max GM: 2500 kg (5512 lb) Engines: Allison 250-C20B TOP per engine: 257 kW (345 SHP) Main transmission: ZF FS 72 B	LBA FAA
	CDN-BS-4 - Same as Variant CBS-4	TCAG

1.6 GENERAL DESCRIPTION OF THE HELICOPTER

The BO 105 is a five-place, multi-purpose, utility helicopter having a hingeless, four-bladed main rotor and a semi-rigid, two-bladed tail rotor. The main and tail rotor blades are constructed of fiber-reinforced composite materials.

Engines

The helicopter is powered by two Allison 250-C20 series engines of the freewheel turbine type. The engine power is transmitted to the main rotor transmission via independent drive systems.

Transmission

The main transmission consists of three major subassemblies in which two bevel gear input stages with freewheel clutches, a spur collector gear stage, and a planetary reduction stage are housed.

Fuel Tanks

The total fuel quantity is stored in three flexible bladder-type tanks, two main tanks (forward and aft) and one supply tank, all installed in the tank compartment beneath the cabin floor.

1.7 HELICOPTER DIMENSIONS

Figure 1-1 provides a three-view drawing of the helicopter with the respective principal exterior dimensions.

Locations on and within the helicopter can be determined in relation to fuselage stations (F.S. or STA.), waterlines (W.L.), and buttock lines (B.L.) measured in millimeters from known reference points.

1.8 CONVERSION CHARTS

See Fig. 1-2 thru 1-7.

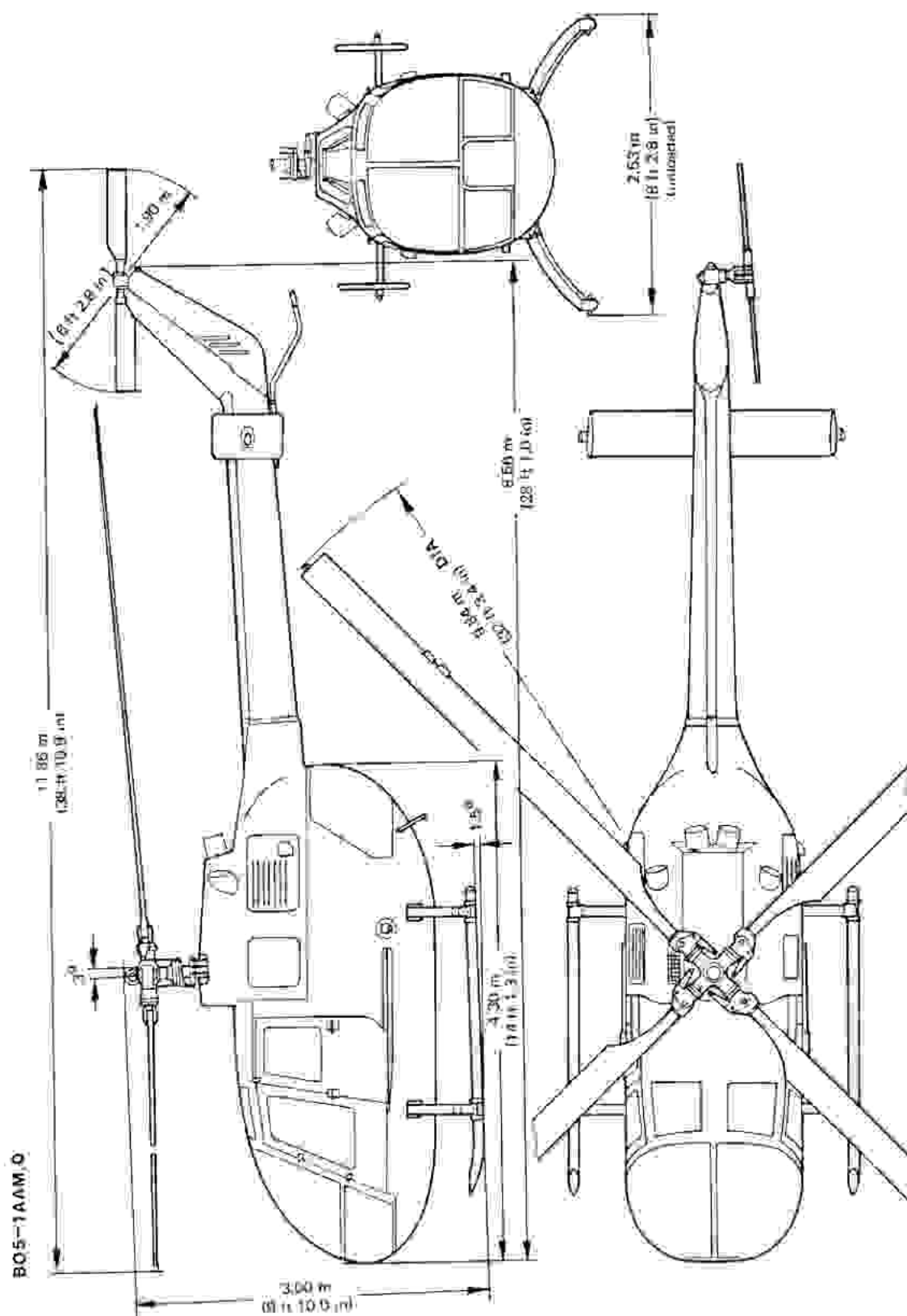


Fig. 1-1 Principal Dimensions

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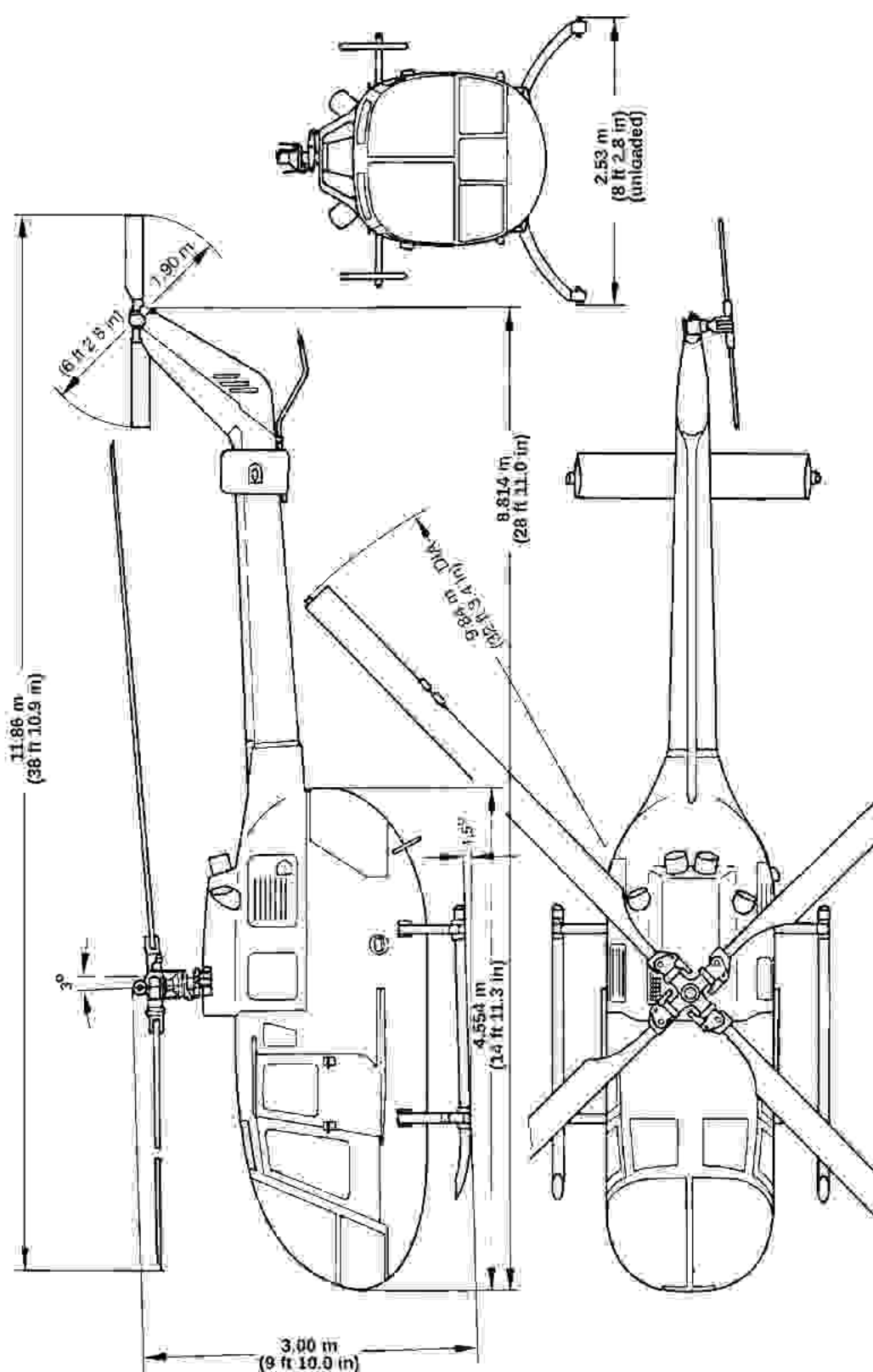


Fig. 1-1 Principal Dimensions

CELSIUS - FAHRENHEIT

$$^{\circ}\text{F} = \frac{9}{5} \times ^{\circ}\text{C} + 32 = 1.8 \, ^{\circ}\text{C} + 17.8$$

$$^{\circ}\text{C} = \frac{5}{9} \times (^{\circ}\text{F} - 32)$$

EXAMPLES: 225 $^{\circ}\text{F}$ = 107 $^{\circ}\text{C}$

845 $^{\circ}\text{C}$ = 1553 $^{\circ}\text{F}$

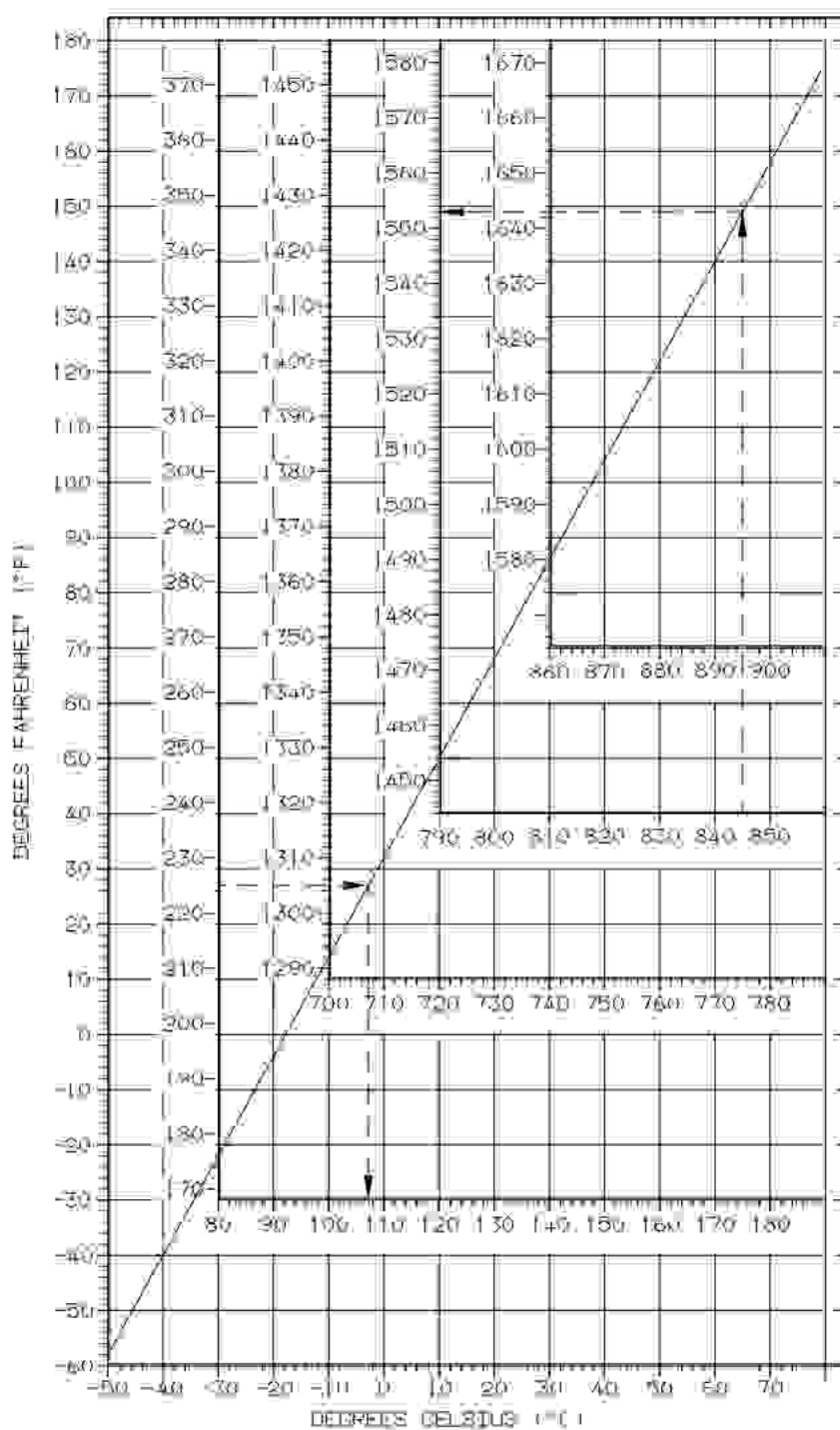


Fig. 1-2 Conversion Chart: Celsius - Fahrenheit

KILOMETERS PER HOUR - KNOTS

$$1 \text{ km/h} = 0.54 \text{ kt}$$

$$1 \text{ kt} = 1.852 \text{ km/h}$$

EXAMPLES: $40 \text{ kt} = 74 \text{ km/h}$
 $200 \text{ km/h} = 108 \text{ kt}$

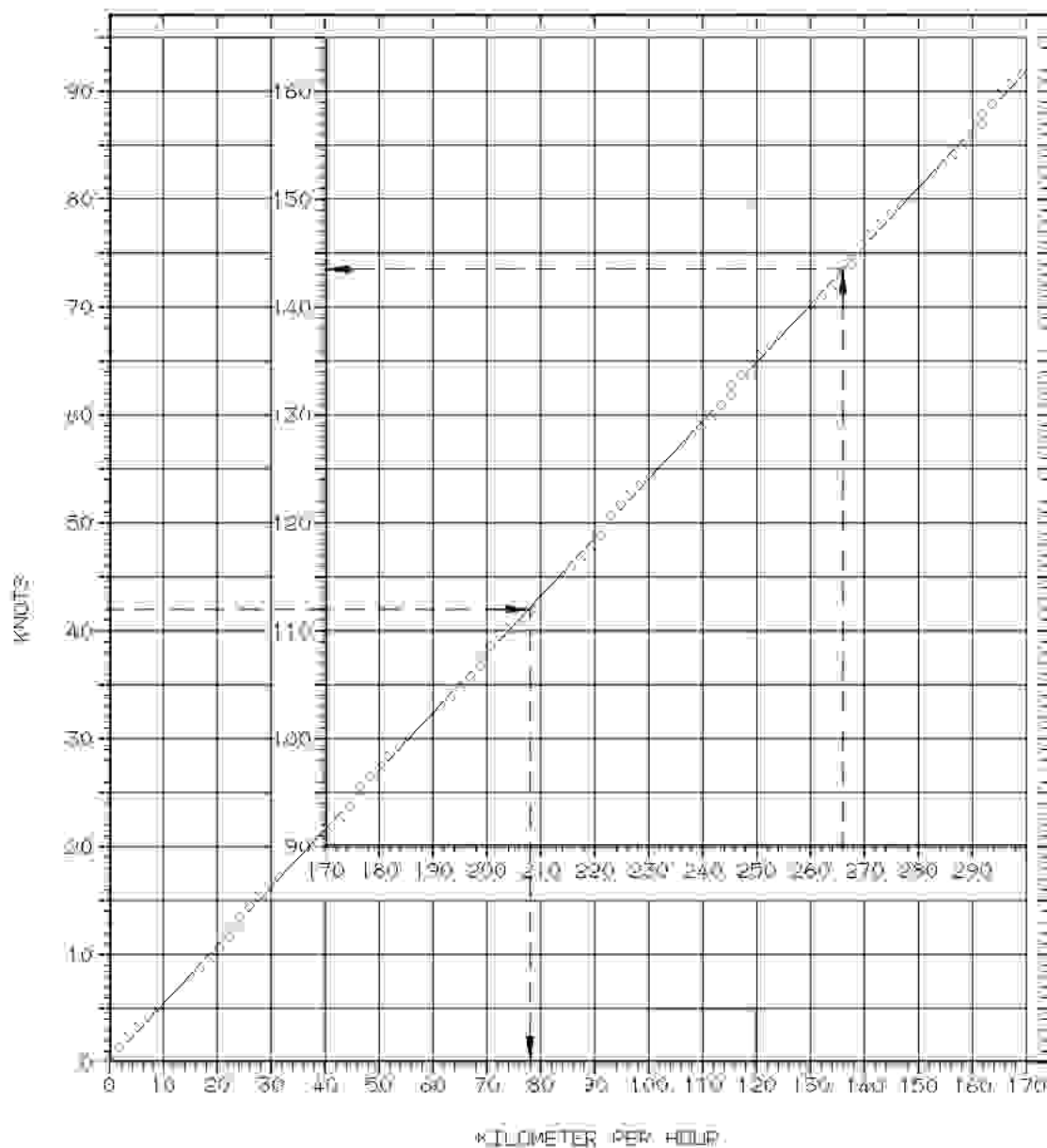


Fig. 1-3 Conversion Chart: Kilometers/Hour - Knots

METERS PER SECOND - FEET PER MINUTE

$$1 \text{ m/s} = 196.85 \text{ ft/min}$$

$$100 \text{ ft/min} = 0.508 \text{ m/s}$$

EXAMPLES: $4.7 \text{ m/s} = 925.2 \text{ ft/min}$
 $340.7 \text{ ft/min} = 17.3 \text{ m/s}$

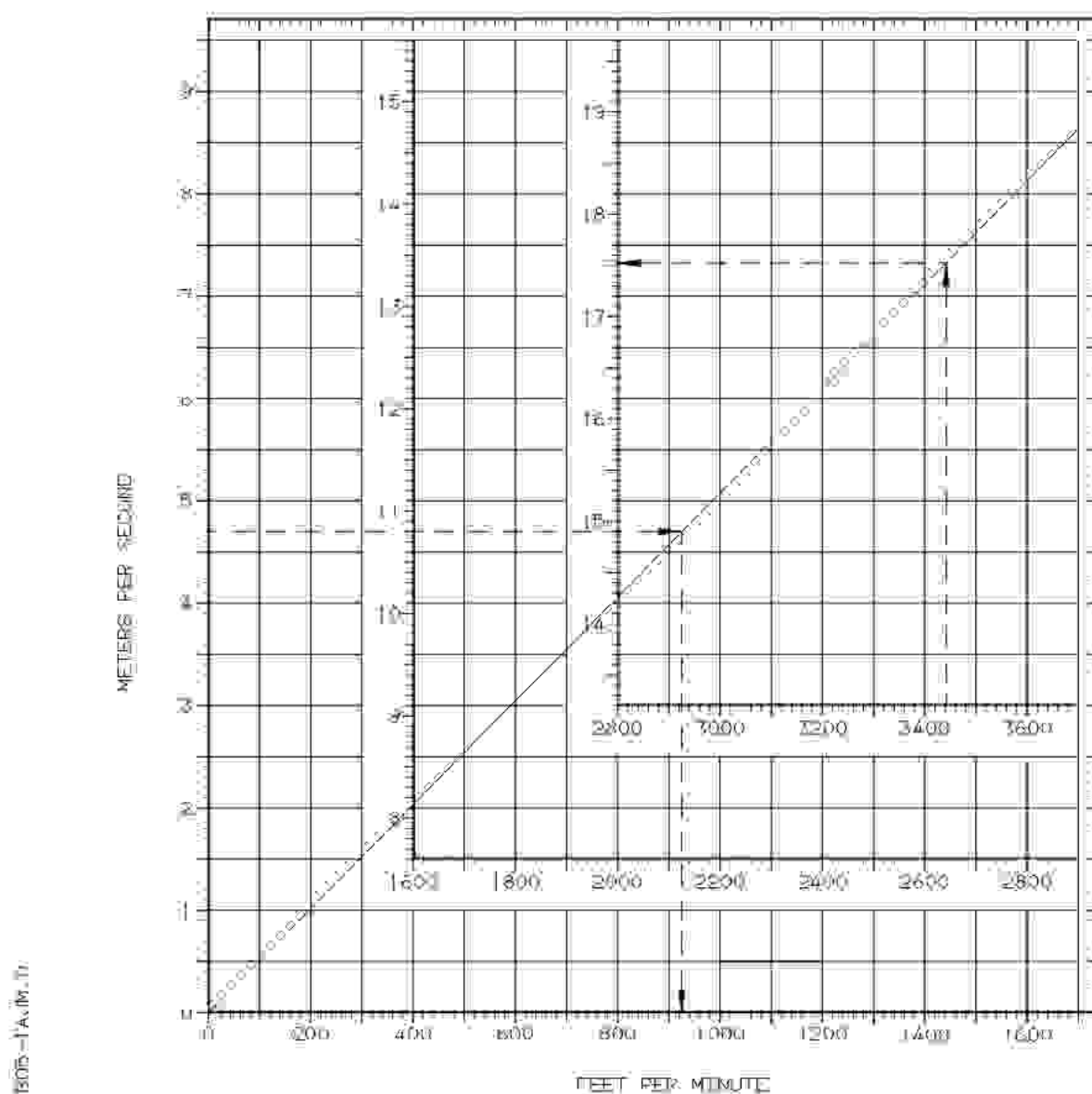


Fig. 1-4 Conversion Chart: Meters/Second - Feet/Minute

KILOGRAMS - POUNDS

$$1 \text{ kg} = 2.205 \text{ lb}$$

$$1 \text{ lb} = 0.454 \text{ kg}$$

EXAMPLES: 540 kg = 1190.7 lb
582 lb = 264.3 kg

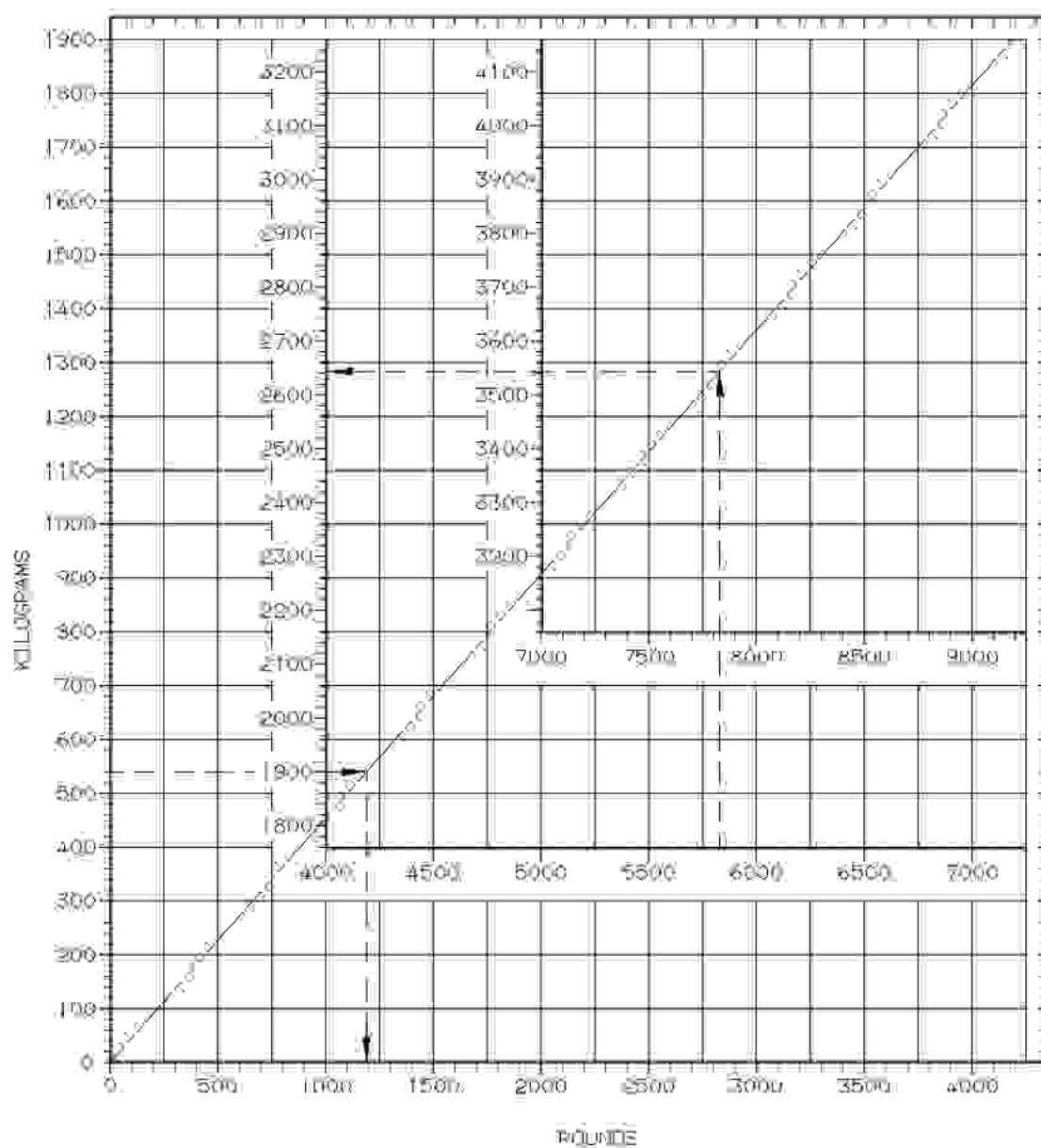


Fig. 1-5 Conversion Chart: Kilograms - Pounds

LITERS - U.S. GALLONS - IMPERIAL GALLONS

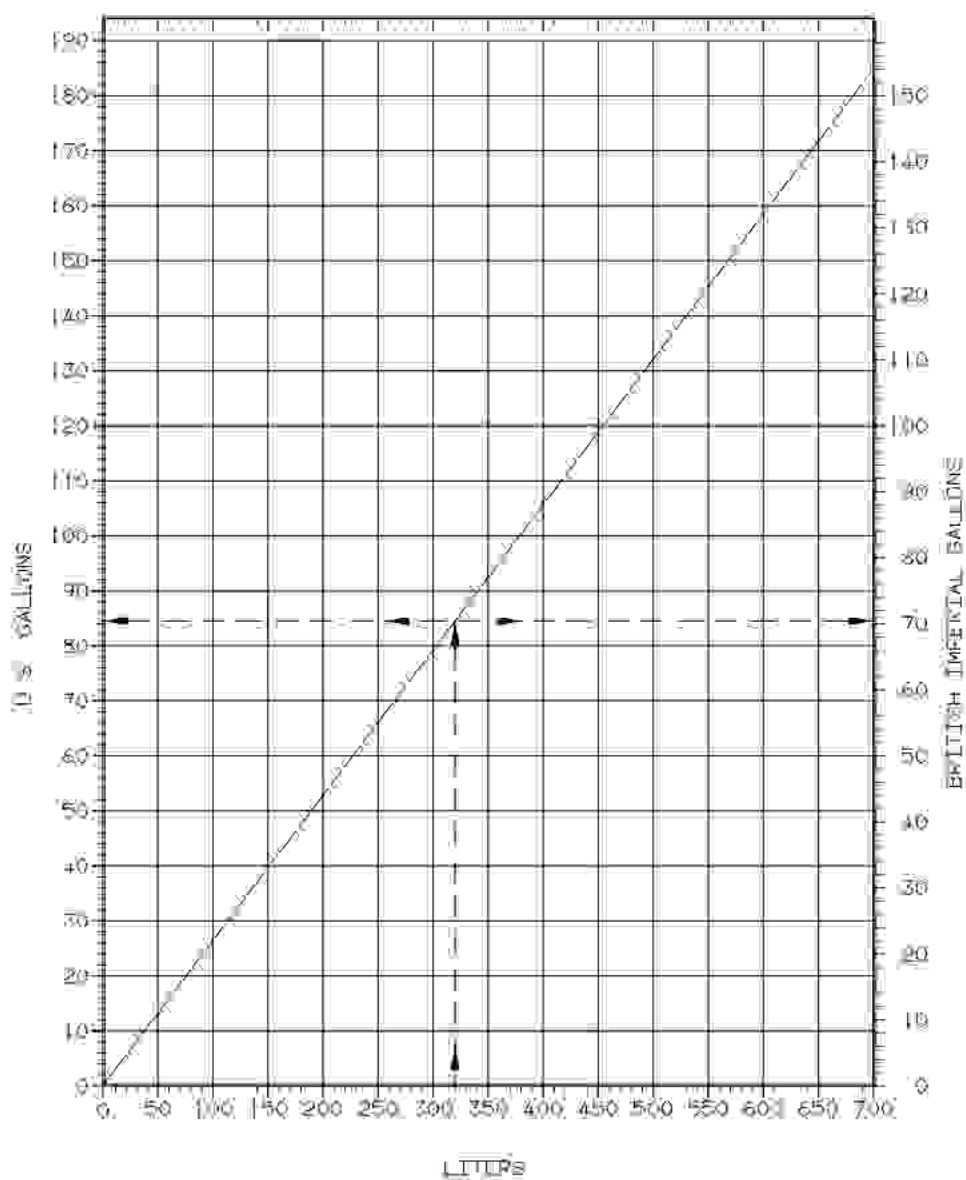


1 U.S. gal	= 3.785 l.
1 l.	= 0.264 U.S. gal



1 Imp. gal	= 4.546 l.
1 l.	= 0.220 Imp. gal

EXAMPLES: 320 l. = 84.5 U.S. gal
 320 l. = 70.4 Imp. gal
 70.4 Imp. gal = 84.5 U.S. gal



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Fig. 1-6 Conversion Chart: Liters - US Gal - Imp. Gal

HECTOPASCALS (MILLIBARS) - INCHES OF MERCURY

$$1 \text{ hPa (mbar)} = 0.02953 \text{ in. Hg}$$

$$1 \text{ in. Hg} = 33.865 \text{ hPa (mbar)}$$

EXAMPLES: $1007.5 \text{ hPa (mbar)} = 29.75 \text{ in. Hg}$
 $30.90 \text{ in. Hg} = 1048.4 \text{ hPa (mbar)}$

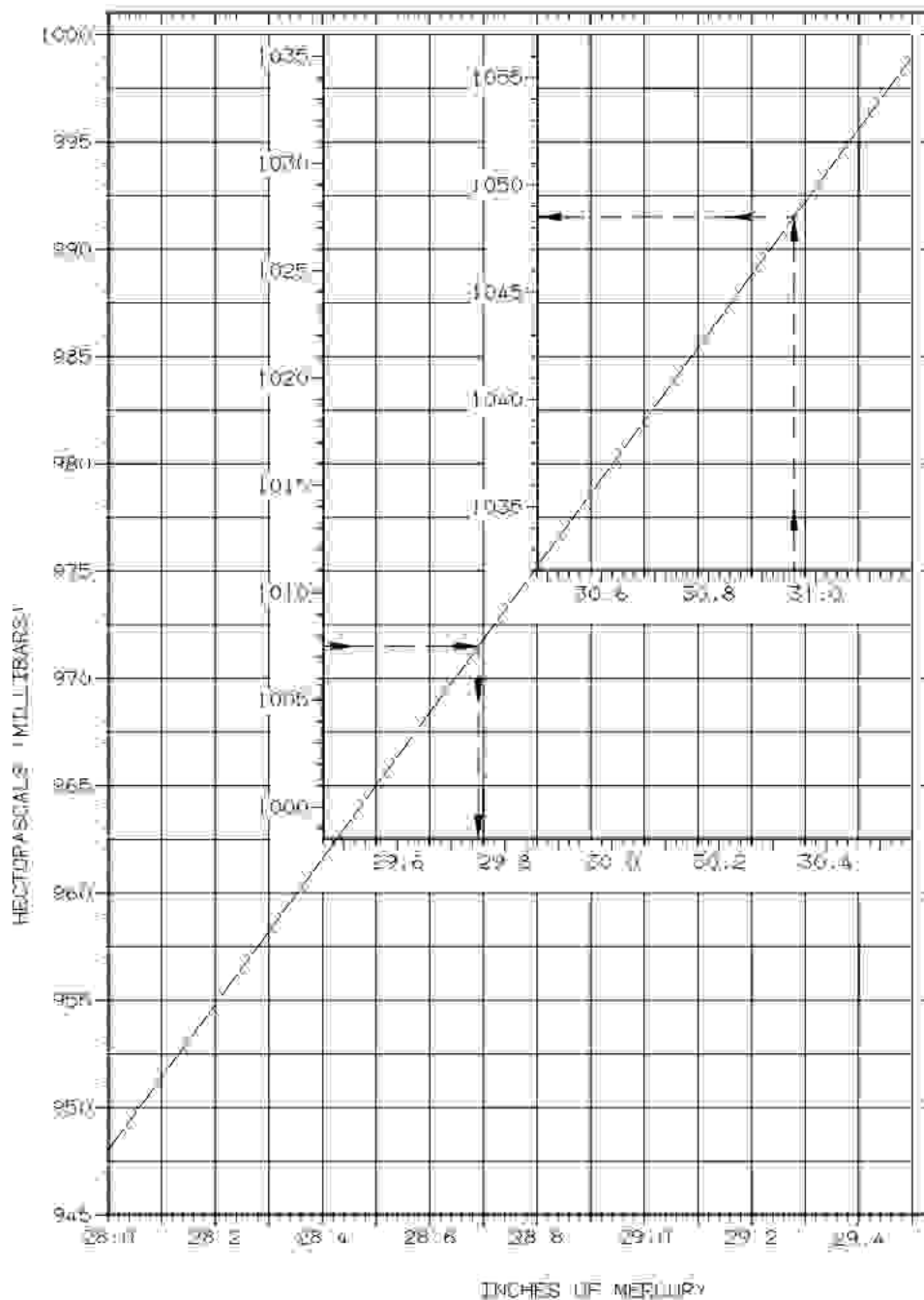


Fig. 1-7 Conversion Chart: Hectopascals - Inches of Mercury

1.9 TERMINOLOGY AND DEFINITIONS OF TERMS

1.9.1 Warnings, Cautions and Notes

Throughout this manual WARNINGS, CAUTIONS and NOTES are used to emphasize important and critical instructions.

WARNING AN OPERATING PROCEDURE, TECHNIQUE, ETC WHICH, IF NOT STRICTLY OBSERVED, COULD RESULT IN PERSONAL INJURY OR LOSS OF LIFE.

CAUTION AN OPERATING PROCEDURE, TECHNIQUE, ETC WHICH, IF NOT STRICTLY OBSERVED, COULD RESULT IN DAMAGE TO OR DESTRUCTION OF EQUIPMENT.

NOTE An operating procedure, technique, condition, etc that is essential to emphasize.

WARNINGS and CAUTIONS immediately precede and are located directly above the text to which they relate.

NOTES follow the text to which they relate and are located directly below the text to which they apply.

1.9.2 Use of Procedural Terms

The procedural term usage and meaning are as follows:

"Shall" and "Must"	have been used to express a mandatory requirement.
"Should"	has been used to express non-mandatory provisions.
"May"	has been used to express permissiveness.
"Will"	has been used only to indicate futurity, never to express a mandatory requirement.

1.10 ABBREVIATIONS AND SYMBOLS

A	- ampere
AC	- alternating current
AEO	- all engines operating
AGL	- above ground level
approx	- approximate
assy	- assembly
ASTM	- American Society for Testing Materials
BAT, BATT	- battery
BO	- Bölkow
CAS	- calibrated airspeed
CFR	- Code of Federal Regulations
CG	- center of gravity
CIS	- Community of Independent States
cm	- centimeter
coll	- collision
COMM	- communication (radio)
compt	- compartment
CTR	- center
cu ft	- cubic feet
DA	- density altitude
DBL	- double
DC	- direct current
e.g.	- for example
EL	- equipment list
EMER	- emergency
ENG	- engine
EPU	- external power unit
EPWR	- emergency power
eqpl	- equipment
etc	- et cetera (and so forth)
EXT	- external, extinguisher
FAA	- Federal Aviation Administration, United States
FAR	- Federal Aviation Regulation
Fig.	- figure
FLM	- flight manual
FMA	- flight manual appendix
FMS	- flight manual supplement
fpm	- feet per minute
ft	- foot (feet)
GAL	- gallon
GEN	- generator
GM	- gross mass
h, hr	- hours of time
Hg	- mercury (hydrargyrum)
HIGE	- hover in ground effect

HOGE	- hover out of ground effect
hPa	- hectopascal
HV	- height velocity
HY, HYD, HYDR	- hydraulic
IAC-AR	- Interstate Aviation Committee - Aviation Register, CIS
IAS	- indicated airspeed
ICS	- intercommunication system
i.e.	- that is (to say)
IFR	- instrument flight rules
IGE	- in ground effect
Imp.	- Imperial
in.	- inch
IND	- indicator
ISA	- International Standard Atmosphere
KCAS	- knots calibrated airspeed
kg	- kilogram
KIAS	- knots indicated airspeed
km	- kilometer
kp	- kilopond
kt	- knot
KTAS	- knots true airspeed
kW	- kilowatt
L	- Left
L, l, ltr	- liter
lb	- pound
LBA	- Luftfahrt-Bundesamt, Federal Republic of Germany
LEP	- list of effective pages
LH	- left hand
m	- meter
max	- maximum, maximal
MCP	- maximum continuous power
MEL	- minimum equipment list
MIL	- military standard
min	- minimum, minimal, minutes of time
MM	- mast moment
mm	- millimeter
MSL	- mean sea level
N ₁ , n1	- gas generator speed
N ₂ , n2	- power turbine speed
NAV	- navigation (radio)
No.	- number
N _{Ro}	- rotor speed
OAT	- outside air temperature
OEI	- one engine inoperative
OGE	- out of ground effect
Pa	- Pascal
PA	- pressure altitude
PAX	- passenger

P/N	- part number
PRESS	- pressure
psig	- pound-force per square inch gage
PU	- polyurethane
PWR	- power
R	- right
R/C	- rate of climb
R/D	- rate of descent
RD	- reference datum
Rev.	- revision
RH	- right hand
RPM	- revolutions per minute
S-	- serial (manufacturer's number)
SB	- service bulletin
sec	- seconds of time
SEL	- selector
SGL	- single
SL	- sea level
S/N	- serial number
sq	- square
STBY	- standby
std	- standard
sw	- switch
T	- transmission
TAS	- true airspeed
TCAG	- Transport Canada Airworthiness Group
TEMP	- temperature
Temp. Rev.	- temporary revision
T/O	- takeoff
TOP	- takeoff power
TOT	- turbine outlet temperature
U.S., US	- United States
V	- volt
VFR	- visual flight rules
VHF	- very high frequency
V _{NE}	- never-exceed speed (velocity never exceed)
VOR	- very high frequency omnirange station
V _Y	- best rate-of-climb speed
XMSN	- transmission
°C	- degrees Celsius (centigrade)
°F	- degrees Fahrenheit
σ	- density ratio
$\frac{1}{\sqrt{\sigma}}$	- true airspeed factor (the reciprocal of the square root of the density ratio, at the density altitude)
ΔPA	- pressure altitude correction